

2 Vitamin A is an essential component of the human diet.

Rice is a staple food in parts of the world where a lot of people have vitamin A deficiency.

Rice contains no β -carotene or vitamin A.

Cells in the human intestine can convert β -carotene from the diet to vitamin A.

Some soil bacteria can produce large quantities of β -carotene.

(a) Describe how genetic engineering could be used to produce rice that contains β -carotene.

(4)

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(b) (i) Describe the possible risks of growing genetically modified rice plants.

(3)

(ii) Suggest how one of these risks could be reduced.

(1)

(Total for Question 2 = 8 marks)

